

Codebook Development History

Cross-codebook iteration summary with H&K- style reporting

Esteban Degetau
World Bank
estebandegetau@gmail.com

Agustín Samano
World Bank
asamanopenaloza@worldbank.org

June 30, 2026

Iteration Timeline

3

CB	Iter	Version	Date	Stage	Model
c1	1	0.1.0	2026-02-20	s1	Haiku
c1	2	0.1.0	2026-02-20	s1	Haiku
c1	3	0.1.0	2026-02-21	s1	Haiku
c1	4	0.2.0	2026-02-28	s1	Qwen 72B
c1	5	0.2.0	2026-03-02	s1	Qwen 72B
c1	6	0.2.0	2026-03-03	s2	Qwen 72B
c1	7	0.2.0	2026-03-03	s2	Llama 70B
c1	8	0.2.0	2026-03-03	s1	Haiku
c1	9	0.2.0	2026-03-03	s2	Haiku
c1	10	0.3.0	2026-03-03	s1	Haiku
c1	11	0.3.0	2026-03-03	s2	Haiku
c1	12	0.4.0	2026-03-04	s2	Haiku
c1	13	0.4.0	2026-03-09	s3	Llama 70B
c1	14	0.4.0	2026-03-09	s3	Llama 70B
c1	15	0.4.0	2026-03-09	s3	Haiku
c1	16	0.4.0	2026-03-10	s3_manual_analysis	Haiku
c1	17	0.5.0	2026-04-04	s1	Haiku
c1	18	0.5.0	2026-04-04	s2	Haiku
c1	19	0.5.0	2026-04-04	s3	Haiku
c1	20	0.5.0	2026-04-04	s3_manual_analysis	Haiku
c1	21	0.6.0	2026-04-05	s1	Haiku
c1	22	0.6.0	2026-04-05	s1	Gemini 2.5 Flash
c1	23	0.6.0	2026-04-05	s1	Gemini 2.5 Flash
c1	24	0.6.0	2026-04-05	s2	Gemini 2.5 Flash
c1	25	0.6.0	2026-04-05	s3	Gemini 2.5 Flash
c1	26	0.6.0	2026-04-06	s2	Haiku
c1	27	0.6.0	2026-04-07	s3	Haiku
c1	28	0.6.0	2026-04-07	s3_manual_analysis	Haiku
c2a	1	0.1.0	2026-04-08	s0	—
c2a	2	0.2.0	2026-04-08	s0	—
c2a	3	0.2.0	2026-04-08	s0	—
c2a	4	0.2.0	2026-04-09	s1	Qwen 72B
c2b	1	0.1.0	2026-04-09	s0_manual_analysis	Gemini 3.1 Flash
c2b	2	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	3	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	4	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	5	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	6	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	7	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	8	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	9	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	10	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	11	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	12	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	13	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	14	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	15	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	16	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	17	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	18	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	19	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	20	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	21	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	22	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	23	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	24	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	25	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	26	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	27	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	28	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	29	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	30	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	31	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	32	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	33	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	34	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	35	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	36	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	37	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	38	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	39	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	40	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	41	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	42	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	43	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	44	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	45	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	46	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	47	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	48	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	49	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	50	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	51	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	52	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	53	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	54	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	55	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	56	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	57	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	58	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	59	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	60	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	61	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	62	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	63	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	64	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	65	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	66	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	67	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	68	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	69	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	70	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	71	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	72	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	73	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	74	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	75	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	76	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	77	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	78	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	79	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	80	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	81	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	82	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	83	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	84	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	85	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	86	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	87	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	88	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	89	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	90	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	91	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	92	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	93	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	94	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	95	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	96	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	97	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	98	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	99	0.1.0	2026-04-09	s0	Gemini 2.5 Flash
c2b	100	0.1.0	2026-04-09	s0	Gemini 2.5 Flash

S1: Behavioral Tests

S1 results for the latest formal (Haiku/Sonnet) run per codebook, following H&K Figure 3.

Figure 1: S1 behavioral test results. For Tests I–III, higher is better. For Test IV, Fleiss' Kappa shown.

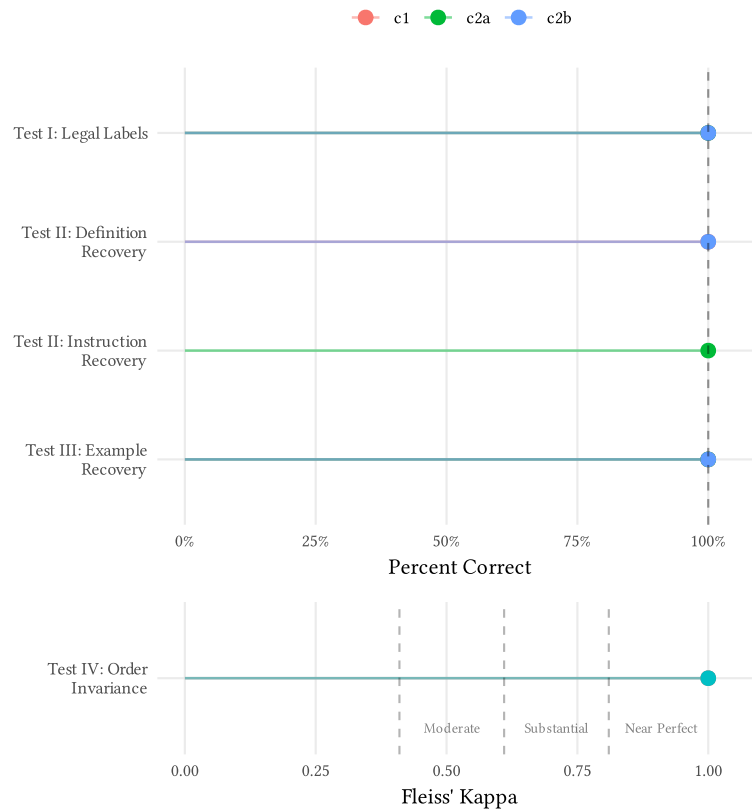


Table 2

CB	Test	Result	Threshold	Status
c1	I: Legal Labels	100%	$\geq 100\%$	Pass
c1	II: Definition Recovery	100%	$\geq 100\%$	Pass
c1	III: Example Recovery	skipped	$\geq 100\%$	Pass
c1	IV: Order Invariance	0.0% (k=1.00)	$< 5\%$	Pass
c2a	I: Legal Labels	100%	$\geq 100\%$	Pass
c2a	II: Definition Recovery	100%	$\geq 100\%$	Pass
c2a	III: Example Recovery	skipped	$\geq 100\%$	Pass
c2a	IV: Order Invariance	skipped	$< 5\%$	Pass
c2b	I: Legal Labels	100%	$\geq 100\%$	Pass
c2b	II: Definition Recovery	100%	$\geq 100\%$	Pass
c2b	III: Example Recovery	skipped	$\geq 100\%$	Pass
c2b	IV: Order Invariance	0.0% (k=1.00)	$< 5\%$	Pass

S2: Zero-Shot Evaluation

C1 Measure Identification

Table 3

C1 S2 (iteration 26, claude-haiku-4-5-20251001)

Metric	Estimate [95% CI]	Target	Status
combined_recall	0.794 [0.765, 0.821]	≥ 0.90	Fail
tier1_recall	0.914 [0.845, 0.975]	≥ 0.95	Fail
precision	0.954 [0.938, 0.970]	≥ 0.70	Pass
tier2_recall	0.782 [0.751, 0.811]	—	—
f1	0.867 [0.848, 0.886]	—	—
accuracy	0.784 [0.757, 0.811]	—	—
specificity	0.710 [0.611, 0.793]	—	—

C2b Motivation Classification

Table 4

C2b S2 (iteration 23, claude-sonnet-4-20250514)

Metric	Estimate [95% CI]	Target	Status
weighted_f1	0.658 [0.511, 0.812]	≥ 0.70	Fail
exogenous_precision	0.750 [0.550, 0.933]	≥ 0.85	Fail
macro_f1	0.659 [0.511, 0.805]	—	—
accuracy	0.658 [0.526, 0.816]	—	—
exogenous_recall	0.682 [0.500, 0.864]	—	—
exogenous_f1	0.714 [0.556, 0.851]	—	—

C2b Per-Class Performance

Table 5

Class	Precision	Recall	F1	N
SPENDING_DRIVEN	0.583	0.700	0.636	10
COUNTERCYCLICAL	0.500	0.500	0.500	6
DEFICIT_DRIVEN	0.857	0.857	0.857	7
LONG_RUN	0.692	0.600	0.643	15

Metric Trajectories

Figure 2: C1 combined recall across formal (Claude) iterations

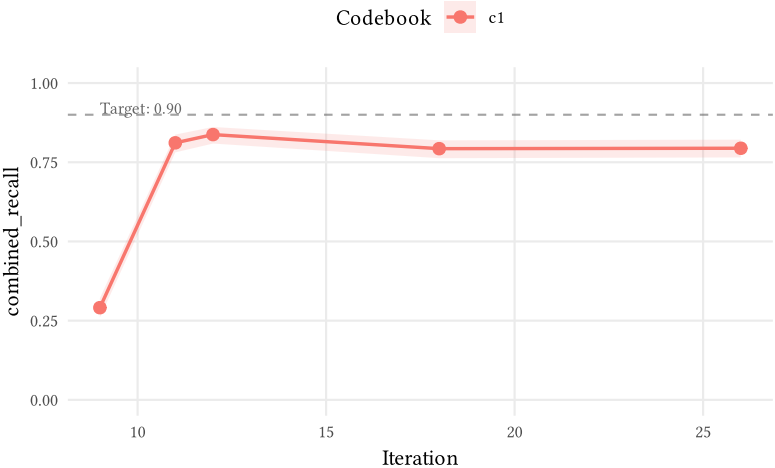


Figure 3: C2b weighted F1 across formal (Claude) iterations

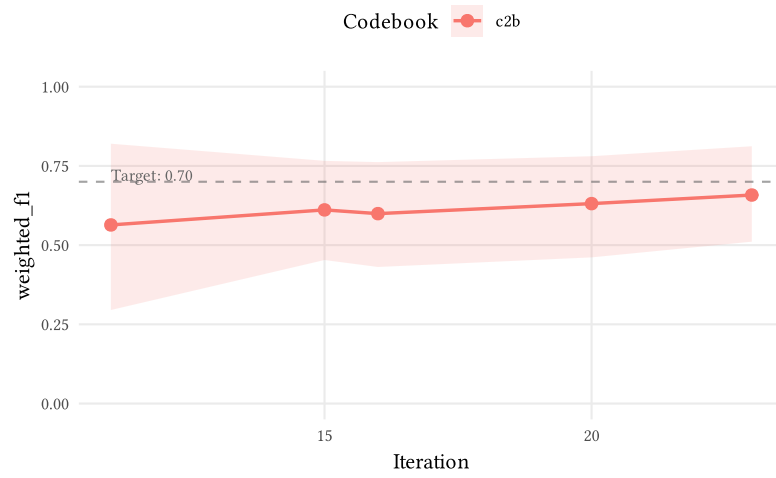
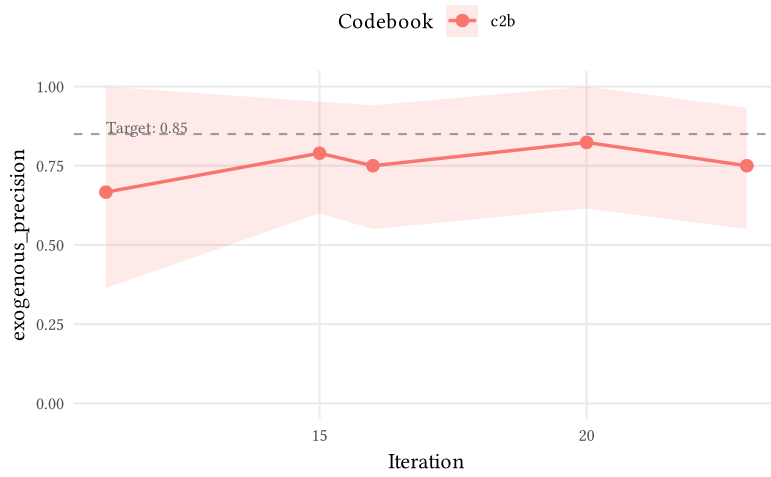


Figure 4: C2b exogenous precision across formal (Claude) iterations



S3: Error Analysis

S3 Behavioral Tests

Figure 5: C1 S3 behavioral tests (latest formal run)

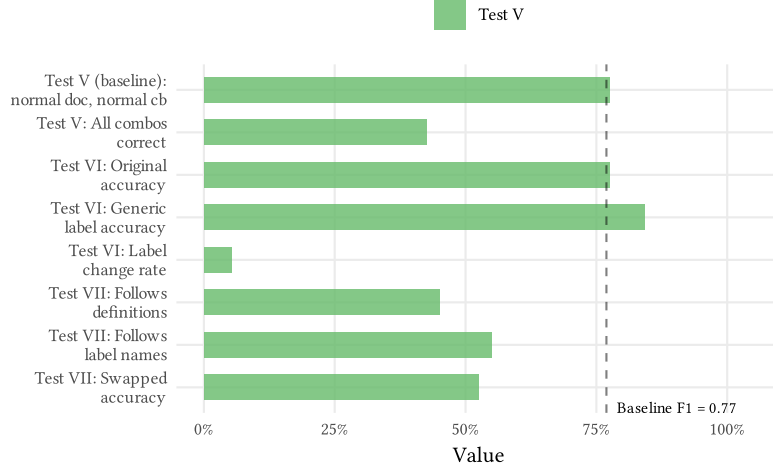
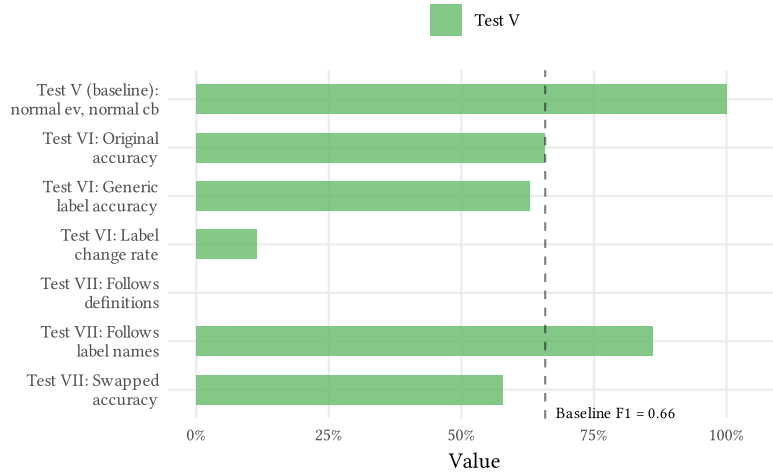


Figure 6: C2b S3 behavioral tests (latest formal run)



Ablation Studies

Table 6: C1 ablation (latest formal run)

Condition	Accuracy	F1	Δ Accuracy	Δ F1
Full codebook	0.775	0.769	—	—
No label definitions	0.775	0.757	0.000	—
No clarifications	0.650	0.708	0.125	—
All removed	0.675	0.723	0.100	—

Table 7: C2b ablation (latest formal run)

Condition	Accuracy	F1	Δ Accuracy	Δ F1
Full codebook	0.658	0.658	—	—
No label definitions	0.579	0.631	0.079	0.027
No clarifications	0.605	0.666	0.053	-0.008
All removed	0.553	0.578	0.105	0.080

Manual Error Analysis

Table 8: C1 manual error analysis (latest)

Category	Count	Proportion
A: LLM correct	31	0.78
B: Incorrect gold standard	6	0.15
C: Document error	0	0.00
D: LLM non-compliance	0	0.00
E: Semantics/reasoning mistake	0	0.00
F: Other	3	0.07

Table 9: C2b manual error analysis

Category	Count	Proportion
A: LLM correct	24	0.62
B: Incorrect gold standard	0	0.00
C: Document error	2	0.05
D: LLM non-compliance	0	0.00
E: Semantics/reasoning mistake	3	0.08
F: Other	10	0.26